

Jovana Kondic

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About Me

I'm a PhD candidate at MIT working at the intersection of **vision-language models**, **synthetic data generation**, and **chart understanding**. I co-**led ChartNet**, an open-source chart-understanding dataset that trained IBM's Granite Vision 4 models to outperform orders-of-magnitude larger models — including Claude Opus 4.6 — at chart summarization and chart-to-table extraction, now trending on Hugging Face with 35K+ downloads and integrated into Docling, IBM's widely-used open-source document-parsing toolkit.

Education

Massachusetts Institute of Technology

PhD in Electrical Engineering and Computer Science

Cambridge, MA

Current - September 2026

- Advised by Prof. Aude Oliva

Massachusetts Institute of Technology

MS in Electrical Engineering and Computer Science

Cambridge, MA

Sept 2021 - January 2024

- Thesis topic: Monte Carlo Methods for Motion Planning and Goal Inference
- Coursework: algorithms for inference, deep reinforcement learning, robotic manipulation, smart control of decarbonized power systems

Princeton University

BSE in Electrical Engineering; certificates in Cognitive Science, Robotics, and ML & Statistics

Princeton, NJ

Sept 2017 - June 2021

- Thesis topic: Towards Socially Aware Robot Learning: Inferring Human Objectives and Latent Safety Preferences from Observations
- Coursework: safety-critical robotics, computational neuroscience, game theory, probability & stochastic systems, animal learning, embedded computing, biomechanics, moral philosophy

Research Experience

Computational Perception and Cognition Lab

Massachusetts Institute of Technology

Cambridge, MA

April 2024 - current

- Vision Language Models (VLMs), synthetic data generation, chart understanding

MetaConscious Group

Massachusetts Institute of Technology

Cambridge, MA

March 2023 - December 2023

- Developed a brain-inspired memory architecture to facilitate long-horizon reasoning in LLM-powered agents in multi-agent environments

Algorithmic Alignment Group

Massachusetts Institute of Technology

Cambridge, MA

September 2021 - February 2023

- Implemented MC sampling algorithms for diversity-aware near-optimal motion planning and trajectory prediction

Safe Robotics Lab

Princeton University

Princeton, NJ

September 2020 - May 2021

- Proposed a framework for 1) multi-objective modeling of human behavior and 2) Bayesian inference of task objectives and risk tolerances in safety-critical human-robot interaction scenarios

Intelligent and Interactive Autonomous Systems Group

Stanford University

Palo Alto, CA

June 2020 - Oct 2020

- Achieved up to +50% performance boost (compared to best baseline) in two-player coordination tasks, by helping develop a modular RL policy network that learns separate representations of task rules and partner-specific conventions

Work Experience

IBM Research/ MIT-IBM Watson AI Labs

Research Intern

Cambridge, MA

June 2024 - August 2026

- Co-**led ChartNet**, a chart understanding research project within the IBM Granite Vision Team, coordinating with 20+ researchers across 5 offices; resulted in an ICCV 2025 workshop publication and a CVPR 2026 publication

Massachusetts Institute of Technology

Teaching Assistant

Cambridge, MA

October 2023 - December 2023

- Collaboratively advised 14 student-led projects at the intersection of deep learning, Bayesian modeling, and neuroscience in course 9.58: *The Science of Intelligence* taught by Prof. Tomaso Poggio and Dr. Brian Cheung

Pillar VC

Venture Research Associate

Boston, MA

May 2023 - September 2023

- Identified and evaluated 8 potential investment opportunities within the applied ML sector through comprehensive market research and close collaboration with senior partners

Skills

Languages	English, Italian, German, Serbian	* denotes limited experience
Programming & Planning Languages	Python, Julia*	
Libraries & Toolboxes	Numpy, PyTorch	
Platforms & Frameworks	Github, HuggingFace, Langchain*	

Publications & Workshop Proceedings

J. Kondic, P. Li, D. Joshi, I. Sanchez, B. Wiesel, S. Abedin, A. Alfassy, E. Schwartz, D. Caraballo, Y. G. Cinar, F. Scheidegger, S. I. Ross, D. K. I. Weidele, H. Hua, E. Arutyunova, R. Herzig, Z. He, Z. Wang, X. Yu, Y. Zhao, S. Jiang, M. Liu, Q. Lin, P. Staar, L. Lastras, A. Oliva, and R. Feris (2026). **ChartNet: A Million-Scale, High-Quality Multimodal Dataset for Robust Chart Understanding**. *CVPR*. [Link to paper](#).

J. Kondic, P. Li, D. Joshi, Z. He, S. Abedin, J. Sun, B. Wiesel, E. Schwartz, A. Nassar, B. Wu, A. Arbelle, A. Oliva, D. Gutfreund, L. Karlinsky, and R. Feris (2025). **ChartGen: Scaling Chart Understanding Via Code-Guided Synthetic Chart Generation**. *ICCV Workshop on Graphic Design Understanding and Generation, ICCV Workshop on Curated Data for Efficient Learning*. [Link to paper](#).

IBM Research Granite Vision Team (**J. Kondic** as Core Contributor) (2025). **Granite Vision: a lightweight, open-source multimodal model for enterprise Intelligence**. arXiv: [2502.09927 \[cs.CV\]](#). [Link to paper](#).

S. Gao, A. Fang, Y. Huang, V. Giunchiglia, A. Noori, J. R. Schwarz, Y. Ektefaie, **J. Kondic**, and M. Zitnik (2024). **Empowering biomedical discovery with AI agents**. *Cell* 187.22, pp. 6125–6151. ISSN: 0092-8674. DOI: <https://doi.org/10.1016/j.cell.2024.09.022>. [Link to paper](#).

K. I. Zhao, M. Naim, **J. Kondic**, M. Ernesto Cortes, J. Ge, S. Luo, G. R. Yang, and A. Ahn (2024). **Lyfe Agents: Generative Agents for Low-Cost Real-Time Social Interactions**. *ArXiv*. [Link to paper](#).

J. Ge, K. Zhao, M. Ernesto Cortes, **J. Kondic**, S. Luo, M. Naim, A. Ahn, and G. R. Yang (2024). **Enhancing Understanding in Generative Agents through Active Inquiring**. *NeurIPS Workshop on Intrinsically-Motivated and Open-Ended Learning*.

T. Zhi-Xuan, **J. Kondic**, S. Slocum, J. B. Tenenbaum, V. K. Mansinghka, and D. Hadfield-Menell (2023). **Bayesian Inverse Motion Planning for Online Goal Inference in Continuous Domains**. *ICRA Workshop on Cognitive Modeling in Robot Learning for Adaptive Human-Robot Interactions*. [Link to paper](#).

A. Shih, A. Sawhney, **J. Kondic**, S. Ermon, and D. Sadigh (2021). **On the Critical Role of Conventions in Adaptive Human-AI Collaboration**. *ICLR*. [Link to paper](#).

Y. Liu, A. G. Domel, S. A. Yousefsani, **J. Kondic**, G. Grant, M. Zeineh, and D. B. Camarillo (2020). **Validation and Comparison of Instrumented Mouthguards for Measuring Head Kinematics and Assessing Brain Deformation in Football Impacts**. *Annals of Biomedical Engineering* 48. [Link to paper](#).

Awards

2025	MIT-IBM Watson AI Lab Graduate Research Assistanship , supporting one year of fully funded doctoral research in AI	MIT
2021	Hewlett Packard Fellowship , merit-based award in the amount of \$90,154	MIT
2021	Sigma Xi Honors Society , inducted to the academic honors society for scientific research	Princeton
2020	Stanford Summer Undergraduate Research Fellowship , selected as one of 16 awardees from a pool of international applicants to participate in a fully-funded summer research program at Stanford	Stanford
2014	Google Science Fair , regional finalist	Google

Media Coverage

2026	MIT researchers teach AI models to interpret charts , feature on the ChartNet project	MIT News
2025	Charting the future of AI, from safer answers to faster thinking , profiled for MIT-IBM Watson AI Lab summer research	MIT News
2023	Thriving Stars at MIT EECS , featured in an interview for MIT EECS Thriving Stars	MIT EECS
2020	Jovana Kondic seeks to model the irrational human brain , profiled as a Stanford Summer Undergraduate Research Fellow	Stanford
2020	Princeton Engineering interview , featured in an interview by Princeton Engineering	Princeton

Leadership

2025	Co-organizer @ ICCV Workshop , co-hosted the First Workshop on Memory and Vision at ICCV 2025	MIT
2023	Executive Board member @ Graduate Application Assistance Program , led operations of departmental initiatives providing application guidance to students from underrepresented backgrounds	MIT

Mentorship

2024 - 26	Graduate Research Mentor , mentored four students on research projects: Jennifer Sun (MIT, undergraduate), Ekaterina Arutyunova (MIT, undergraduate), Isaac Sanchez (MIT, undergraduate), and Ahmetcan Yavuz (ETH Zurich, masters)	MIT
2020	Teaching Assistant @ Princeton EE/ Keller Center , collaborated with faculty to implement innovative teaching methods resulting in improved student retention and success	Princeton
2020	Peer Mentor @ Computer Science Council , guided first-year students through curriculum planning and academic support	Princeton
2019	Mentor @ Society of Women Engineers , engaged in panel discussions aimed at inspiring high school students to consider careers in STEM fields	Princeton